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Short Communication

Seroepidemiology of hepatitis B and C virus infections among blood donors in Ethiopia†

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ABSTRACT

This study was conducted to investigate the prevalence of hepatitis B virus (HBV) and hepatitis C virus (HCV) infections among blood donors in different regions in Ethiopia. A total of 56,885 sera were tested for HBsAg and anti-HCV antibodies. Of these, 3.9% were found HBsAg-positive, 0.52% anti-HCV-positive and 0.054% dual positive. HBV prevalence was relatively higher in Adama (5.91%) than Gondar (4.05%), Jimma (3.87%), Addis Ababa (3.75%), and Tigray (3.7%); and in males (4.64%) than females (2.1%). Overall, HBV and HCV prevalence increased with age. In conclusion, HBV and HCV seroprevalence among blood donors in Ethiopia is intermediate and low, respectively. This article is protected by copyright. All rights reserved

Keywords: HBV; HCV; Seroepidemiology; Ethiopia

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INTRODUCTION

Hepatitis B virus (HBV) and hepatitis C virus (HCV) are a major public health problem in all parts of the world. Worldwide, over 240 million people are infected with HBV and 185 million with HCV [Mohd Hanafiah et al., 2013; Schweitzer et al., 2015]. HBV and HCV are the leading cause of infectious disease-related morbidity and mortality. According to a recent study by Stanaway *et al*., the annual number of deaths attributable to viral hepatitis rose by 63% from 1990 to 2013 — from 0.89 million to 1.4 million. Of these, HBV and HCV accounted for 96% of hepatitis-related mortality in 2013 alone, with the highest mortality rates seen in Africa and Asia [Stanaway et al., 2016].

Although HBV and HCV infections are a global health problem, there is a distinct geographical variation in their prevalence, with regions having high (≥ 8%), intermediate (2-7%) and low (< 2%) endemicity. Africa is among the highly endemic regions, with an HBV prevalence rate over 8%. Overall, this continent shares 30% of the global HBV burden, with over 75 million chronic carriers [Schweitzer et al., 2015]. Moreover, Africa has the highest HCV burden with a prevalence of 5.3% and an estimated 32 million chronic carriers [Mohd Hanafiah et al., 2013]. Like most African countries, Ethiopia has been categorized among the highly endemic areas for viral hepatitis. However, due to the lack of a nationwide survey and readily available routine serological tests for patients, the exact prevalence of these viruses in Ethiopia remains unknown. Although a few studies have been performed, these were mainly either restricted to subjects from risk groups, HIV patients, a small sample size [Abebe et al., 2003; Bisetegen et al., 2016; Gebreegziabher et al., 2016; Shimelis et al., 2008] or done several years ago [Kefene et al., 1988]. Hence, large-scale studies are needed to monitor the up-to-date prevalence of these viruses and to implement appropriate preventive measures. In this multicentric cross-sectional study, we aimed to determine the prevalence of HBV and HCV infections among

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healthy blood donors to gain insight into the burden of these viruses in the general population of Ethiopia.

MATERIALS AND METHODS

In a cross-sectional study, a total of 56,885 serum samples was collected from healthy blood donors living in five distinct geographical regions in Ethiopia. Between March 2013 and April 2014, 46,052 samples were collected from Addis Ababa, 3,488 samples from Adama, 2,986 samples from Gondar, 465 samples from Jimma and 3,894 samples from Tigray regions. Serum sample collections were carried out at laboratories in national and regional blood banks. Blood donors who fulfilled the national blood bank criteria to be eligible for donation were included. Those who did not meet the inclusion criteria: age < 18 or > 65 years, history of long-term medication use, family history of liver disease, surgery, repeated blood donation < 1 year and pregnant women were excluded from the study. A blood sample of 10 ml was collected from each donor using a sterile BD vacutainer, and the serum was separated by centrifugation and placed in sterile serum storage vials. All samples were screened for hepatitis B surface antigen (HBsAg) and anti-HCV antibodies by using a fourth-generation Enzyme-linked immunosorbent assay (ELISA) kits (DIALB Diagnostics GmbH, Vienna, Austria), according to the manufacturer's instructions. The study was approved by the National Research Ethics Review Committee at the Federal Ministry of Science and Technology, Ethiopia.

RESULTS

Of the 56,885 blood donors tested in this study, 40,204 (70%) were males and 16,681 (30%) were females, aged 18-62 years. The prevalence of HBV, HCV and HBV/HCV coinfection is shown in Table 1. Of the 56,885 samples screened, 2,217 were HBsAg positive, giving the overall nationwide

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HBV prevalence of 3.9%. When we compare each region, HBV prevalence was relatively high in Adama (5.91%) followed by Gondar (4.05%), Jimma (3.87%), Addis Ababa (3.75%) and Tigray (3.7%). With regard to gender, there was a significant difference between males and females in HBV seropositivity. HBsAg prevalence was roughly two times higher in males than females (4.64% vs. 2.1%).

Notably, a very low prevalence of anti-HCV antibodies was found in the present study. The overall prevalence of HCV was 0.52%, corresponding to 294 anti-HCV positive individuals out of 56,885 tested. No difference was observed in anti-HCV positivity between males and females. Regarding the prevalence of HBV/HCV coinfection, the presence of both HBsAg and anti-HCV antibodies were seen in 31 out of 56,885 (0.054%) individuals. Of these, 23 were males and 8 were females.

Age-related prevalence of HBV and HCV was also assessed in this study. As shown in Figure 1 the prevalence of HBV and HCV increases with age with a peak in the age group 40-49 (5.33% for HBV and 0.81% for HCV). This peak is then followed by a gradual decrease in prevalence in the older group > 50 years old (to 4.57% for HBV and to 0.78% for HCV).

DISCUSSION

Information on the epidemiology of viral hepatitis in Ethiopia is still lacking. To the best of our knowledge, this is the first large-scale nationwide epidemiological study of HCV and HBV infections from Ethiopia. Herein we report the prevalence of HBV and HCV in different geographical regions—Addis Ababa, Adama, Gondar, Jimma, and Tigray. These regions are approximately 100 to 1,500 km distant from each other and differ in their socioeconomic, culture and ethnicity. The overall prevalence of HBsAg was 3.9%, and anti-HCV was 0.52%. These prevalence rates classify the country as an

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intermediate (2–7%) endemic area for HBV infection and low (< 1.5%) for HCV [Mohd Hanafiah et al., 2013; Schweitzer et al., 2015].

Region wise HBsAg positivity rates ranged from 3.7% in the northern Tigray region to 5.91% the eastern Adama region. The centrally located Addis Ababa city (capital city) had HBsAg prevalence of 3.75%, whereas the southwestern region Jimma had 3.87%. These results are in concurrence with previous studies reported from Addis Ababa and Gondar [Abebe et al., 2003; Shimelis et al., 2008; Tessema et al., 2010], but lower than those studies from Jijiga [Mohammed and Bekele, 2016], Sekota [Gebreegziabher et al., 2016], Wolaita Sodo [Bisetegen et al., 2016] regions, and a nationwide study (10.8%) [Kefene et al., 1988]. This may be attributable to the difference in the study period, study subjects, and sample size. For instance, the previous nationwide study was conducted three decades ago and the sample size was limited to 5270 young males. Whereas the majority of study subjects at Jijiga blood banks were replacement donors [Mohammed and Bekele, 2016] and those studies by Gebreegziabher et al. and Bistetegen et al. were limited to 482 and 390 samples, respectively, included risk groups and suspected patients [Gebreegziabher et al., 2016], our findings were drawn from a large sample size consisting of only voluntary donors.

The HBsAg prevalence among blood donors in this study is comparable with reports from neighboring countries; 4.2% in Sudan [Abdo et al., 2015] and 2.58% in Eritrea [Fessehaye et al., 2011], but lower than other Central and Western African countries [Babanejad et al., 2016]. Overall, blood donors showed lower HBsAg prevalence compared to the general population [Baha et al., 2013; Behal et al., 2008; Yong-Lin et al., 2012; Zou et al., 2009] although a high prevalence of HBsAg among blood donors was reported from some regions. This might be due to exclusion of risk groups with clinical signs and symptoms of hepatitis in blood donors.

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Regarding the HCV infection, the overall prevalence in the present study was 0.52%, which is significantly lower than the previously estimated 2.0% [Mohd Hanafiah et al., 2013]. Interestingly, the prevalence rate was constantly lower than 1% across all regions. Our result is in line with HCV prevalence among blood donors in Southeast African region (0.43%) but lower than Central (2.61%) and West (3.18) Africa regions [Rao et al., 2015]. It is noteworthy that HBV/HCV coinfection is not uncommon in endemic areas because of the shared modes of transmission. Many studies have reported HBV/HCV coinfection prevalence rates, ranging from 0.7 to 15%, as recently reviewed by Konstantinou and Deutsch [Konstantinou and Deutsch, 2015]. In our study, we found only 31 (0.054%) coinfections out of 56, 885 samples tested, which is significantly lower than the aforementioned studies. This might be due to differences in study subjects as HBV/HCV coinfection is more frequently found in high-risk populations, including injecting drug users, hemodialysis patients, organ transplant patients, HIV-positive and β-thalassemia patients [Konstantinou and Deutsch, 2015], whereas our study subjects are healthy blood donors.

Another important observation in this study was gender-related differences in the prevalence of HBV infection. The prevalence of HBV infection was significantly higher in males than in females. This was comparable with findings reported elsewhere [Abebe et al., 2003; Baha et al., 2013; Yong-Lin et al., 2012]. This difference might be due to the higher exposure to occupational HBV risk factors in males than females, or because females are more likely than males to produce hepatitis B surface antibody (anti-HBs) in response to HBV infection [Blumberg, 1979]. Nevertheless, no difference was observed in anti-HCV positivity between males and females.

With regard to age-related analysis, the prevalence of both HBV and HCV infection increased with age from 2.35% to 5.33% for HBV and from 0.34% to 0.81% for HCV. The highest prevalence had been seen among persons 40–49 years of age. Whereas, in the older group over 50 years, the prevalence of

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both viruses decreased progressively. This could be probably due to deaths caused by chronic hepatitis-related cirrhosis and hepatocellular carcinoma in this age group [Yuen et al., 2009]. Moreover, 25% of people who acquire HBV as children will develop primary liver cancer or cirrhosis as adults. This is the case in sub-Saharan Africa, where transmission predominantly occurs in infants and children by perinatal and horizontal routes [Schweitzer et al., 2015]. Thus, it is important to note that many people are at high risk of developing viral hepatitis-related chronic liver disease at the later age, due to the fact that antiviral therapy is currently not available in Ethiopia.

In conclusion, this study shows that the prevalence rate of HBV is intermediate and that of HCV is low in Ethiopia and their prevalence more or less similar across the different regions. Although these findings are useful for making estimates about the overall viral hepatitis burden in Ethiopia, it might underestimate the prevalence in the general population due to exclusion of risk groups with clinical signs and symptoms of hepatitis in blood donors. Thus, we strongly encourage a need for national-level surveillance and public campaigns to raise awareness.

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Figure legend

Figure 1. Age-specific prevalence of HBsAg and anti-HCV antibody among blood donors

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Table 1. Prevalence of HBV and HCV infection and coinfection by region and gender

	No. Tested	HBsAg positive N (%)	Anti-HCV-Ab positive* N (%)	Coinfection N (%)†	Total N (%)
Region					
Addis Ababa	46052	1728 (3.75)	248 (0.54)	29 (0.063)	2005 (4.35)
Adama	3488	206 (5.91)	13 (0.37)	0	219 (6.28)
Gondar	2986	121 (4.05)	15 (0.50)	1 (0.033)	137 (4.59)
Jimma	465	18 (3.87)	3 (0.65)	0	21 (4.52)
Tigray	3894	144 (3.7)	15 (0.39)	1 (0.026)	160 (4.11)
Total	56885	2217 (3.9)	294 (0.52)	31 (0.054)	2542 (4.47)
Gender					
Male	40204	1866 (4.64)	196 (0.49)	23 (0.06)	2085 (5.19)
Female	16681	351 (2.1)	98 (0.59)	8 (0.05)	457 (2.74)
Total	56885	2217 (3.9)	294 (0.52)	31 (0.054)	2542 (4.47)

† Positive for both HBsAg and Anti-HCV antibody

*Ab, antibody

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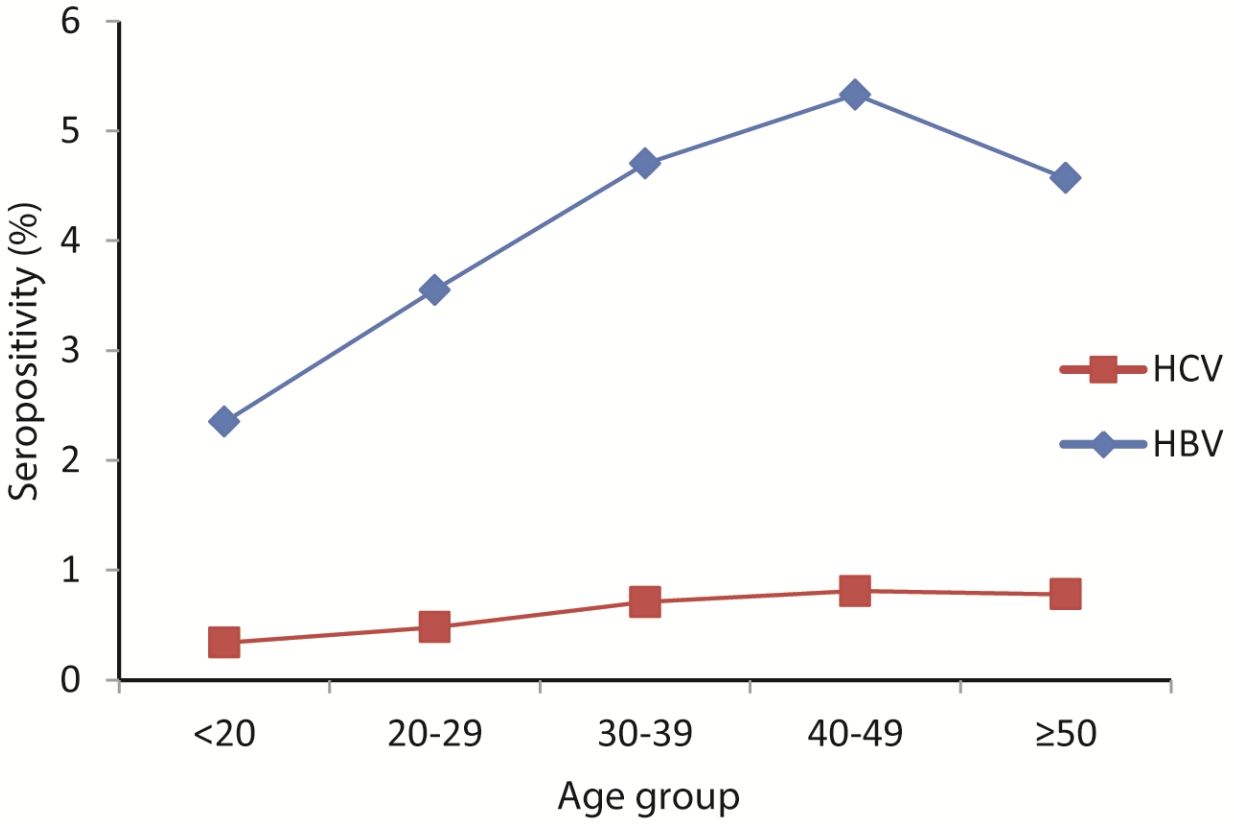


Figure 1

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